Post-Visual Residual Key–Value Test-Time Tuning for Event-Adaptive Remote-Sensing Damage Assessment

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***Abstract*—Building-damage assessment from paired pre-/post-event satellite imagery is high-stakes, yet every newly unfolding disaster is an implicit distribution shift for any fixed model. We propose Post-Visual Residual *Key–Value Test-Time Tuning* (KV-TTT), a weight-disjoint, budget-bounded adaptation that operates exclusively in the activations of a frozen source VLM. Given a tiny balanced support set collected after the event, KV-TTT (1) back-propagates a label-token loss through a frozen decoder and collects "correctness gradients" on the *post-event* image tokens for a few k/v-projection outputs; (2) assembles their covariance into a low-rank corrective subspace $\boldsymbol{B}$; (3) injects a masked, bounded residual $\boldsymbol{Z}' = \boldsymbol{Z} + \boldsymbol{M} \odot \big((\boldsymbol{Z}\boldsymbol{B})\,\mathrm{diag}(\alpha_{\max}\tanh \boldsymbol{a})\,\boldsymbol{B}^\top\big)$ on those rows; and (4) fits the 64 bounded scalars $\boldsymbol{a}$ with four full-batch Adam steps over the support. We report leave-one-event-out results on BRIGHT ($244{,}976$ instances): on every fold completed so far KV-TTT improves macro-F1 and negative log-likelihood, is stable across three source seeds, and beats the weight-space baseline that fine-tunes the same twelve support instances into the LoRA (which overfits and collapses the model). A cross-event variant that reuses a subspace from other disasters performs on par with the online one. No weight is touched; only the residual is moved.**

***Index Terms*—test-time adaptation, building damage assessment, remote sensing, vision–language models, key–value memory.**

I. Introduction

Building-damage assessment is one of the first and most decision-relevant signals available after a disaster: for every building footprint in a populated area the analyst needs to know whether it is *intact*, *damaged* or *destroyed*, using a pre-event reference and a freshly acquired post-event image. Automating it matters precisely because the surveying institutions themselves are often incapacitated at the moment the answer is needed.

Vision–language models (VLMs) are strong candidates: they carry rich visual priors [1], produce calibrated per-class scores, and adapt well to remote sensing via low-rank fine-tuning [2], [3]. Yet every new disaster event is a distribution shift in disguise — different sensor, geometry, lighting, land cover, and damage semantics. A model that performs well on source data degrades, sometimes abruptly, on the event it is deployed to first. The conventional remedy—weight fine-tuning even at low rank—is brittle when the adaptation budget is a dozen samples: it memorizes the support and collapses confidence. In our measurements, updating the source LoRA on exactly the twelve support examples that KV-TTT also sees drives macro-F1 on the target from 0.387 to 0.340, pushes destroyed-signal recall from 0.70 to 0.12, and makes the model mass probability on a single damage class.

We take a different route. A transformer is a deep key–value architecture [4]: each decoder layer projects its context into keys and values that feed attention [5], [6]. The *weights* encode the general prior; the *activations* of the k/v projections carry instance-specific evidence about the post-event change. Our key observation is that, at support time, the label-token loss induces a concentrated, low-rank gradient exactly on the post-event image-token rows of those projections. We collect these "correctness gradients", summarize them into a low-rank subspace $\boldsymbol{B}$, and fit a handful of bounded scalar gates that reshape only those token rows in k/v output space. The model, its LoRA, its text tokens, and its pre-event image tokens never change. The result is adaptation that is weight-disjoint, bounded between identity and a maximal gate (at zero coefficients the frozen model is bit-exact), deterministic (full-batch, no sampling), and cheap (one forward pass per query at inference). Figure 1 and Table I report leave-one-event-out isolation on BRIGHT [7].

II. Related Work

Test-time adaptation. Test-time training [8], TENT [9], and their online/continual extensions [10] modify normalization statistics or the training objective at inference, usually assuming an unlabelled test stream and changing per-sample feature statistics. For VLMs, test-time prompt tuning [11] optimizes a soft prompt on unlabelled test batches; this needs a batch of correlated samples and still changes a weight-like parameter at query time. Applying these to a frozen generative VLM is awkward because the "single sample" is an entire sequence. KV-TTT instead performs a small, deterministic parametric correction in the key–value manifold, conditioned on a tiny but labelled event support, and leaves the forward path untouched at query time. **Transformers as key–value memories.** Decoder layers can be read as interpretable key–value stores [5], [12]; this view motivated memory editing [12], [13], few-shot in-context learning [6], and prefix-prompt control [14]. Unlike full KV-retrieval or ICL, KV-TTT edits a *direction* (a subspace) we control, keeps every weight untouched, and pays none of the per-query context-length cost of ICL. **Parameter-efficient adaptation.** LoRA [2], adapters [15], and prompt-learning [14] change weights at the source stage, amortizing one model over a task family but offering no way to adapt at deployment without re-fitting weights on a tiny budget. Our controller is an orthogonal complement: it plugs into any trained LoRA at the forward hook level. **Remote sensing for damage.** xView2 [16] and the humanitarian BRIGHT [7] benchmarks underpin recent VLMs for damage grading [17], building on remote-sensing foundation-model features [18]. Few-shot, test-time specialization of such models with cross-event transfer has essentially not been evaluated; our protocol gives a drop-in for any LoRA-ready VLM.

III. Method

A. Setup and notation

Let f_θ be a frozen source model: a VLM whose LoRA adapter was fine-tuned on source events. Its input packs the pre-event image crop, the post-event image crop, an instruction, and an answer span. We write $\mathbf{Z}^{(l,t)} \in \mathbb{R}^{T_{\text{seq}} \times d}$ for the output of the k- or v-projection of decoder layer l at all sequence positions, and $\boldsymbol{M} \in \{0,1\}^{T_{\text{seq}}}$ for the mask that is 1 only on the post-event image tokens (the second visual group). Inputs are 448×448 crops; tokens are arranged as pre-event group, post-event group, instruction, answer.

B. Corrective subspace from correctness gradients

Given the support $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^{12}$, balanced per class, with the model frozen we compute, for selected projections $(l, k/v)$, the label-token loss $\mathcal{L}_i = -\log p_\theta(y_i \mid x_i)$ and

$$G_i^{(l)} = \nabla_{[\mathbf{Z}^{(l)}]_M} \mathcal{L}_i \in \mathbb{R}^{m \times d}, \qquad C^{(l)} = \sum_i (G_i^{(l)})^\top G_i^{(l)},$$
$$B^{(l)} = \operatorname{eigvec}_r \left(C^{(l)}\right) \in \mathbb{R}^{d \times r}. \tag{1}$$

where m is the number of masked post-event tokens and r=16. $B^{(l)}$ spans the dominant right-singular directions of the stacked gradient matrix over the support; we call it the *corrective subspace*. Because $\nabla_{[\mathbf{Z}^{(l)}]_M}$ records, per basis direction of the post-event activation space, the direction in which the label-token log-likelihood most improves for all support examples at once, $C^{(l)}$ is a correlation of local "what would make the answer more correct" signals; G_i across the 12 examples, only the top-r subspace is stable enough to survive the tiny support, the rest is discarded.

C. Post-visual residual controller

For each selected projection we wrap the output with the gated residual

$$\mathbf{Z}'^{(l)} = \mathbf{Z}^{(l)} + \boldsymbol{M} \odot \Big((\mathbf{Z}^{(l)} B^{(l)}) \operatorname{diag}(\alpha_{\max} \tanh a^{(l)}) (B^{(l)})^\top \Big), \tag{2}$$

with $\alpha_{\max}$=0.5 and $a^{(l)} \in \mathbb{R}^r$. The row-masked $\odot$ restricts the disturbance to the *post-event* token rows; at a=0 the transform is the identity, so the source is bit-exact. The controller is a forward hook on the frozen projection; it never touches the weight tensor or the LoRA delta. Selected pairs are the key and value projections of the last two decoder layers ($2 \times 2 = 4$ projections, $4 \times r$=$16 = 64$ scalar gates).

Algorithm 1 KV-TTT (given frozen source LoRA)

Require: Support $\mathcal{S}$ (12 ex.); projections $\{(l,t)\}$; $T, \text{lr}, \lambda, \alpha_{\max}$
Ensure: Controller coefficients a

```
Freeze f_θ; disable caching.
for i ∈ S do
    L_i ← label-span cross-entropy (forward)
    backward; collect G_i^(l) = ∇_[Z^(l)]_M L_i via hooks
    C^(l) ← C^(l) + (G_i^(l))^T G_i^(l); zero-grad
end for
B^(l) ← top-r eigenvectors of C^(l) (eigh)
a ← 0
for t in 1...T do
    ∇ ← Σ_i [∇_a L_i(Z'(a); k, y_i) + 2λa]
    a ← a + Adam(∇, 1)
end for
return {B^(l), a^(l)}
```

D. Test-time fitting

The 64 scalars are fit by full-batch gradient with a tanh scale and weight decay:

$$a^* = \arg\min_a \sum_{i \in \mathcal{S}} \mathcal{L}_i\big(\mathbf{Z}'(a);\, x_i, y_i\big) + \lambda \|a\|^2,$$
$$T{=}4,\ \text{lr}{=}0.05,\ \lambda{=}10^{-3}, \tag{3}$$

each of the T=4 steps accumulates all twelve support examples, then applies Adam with unit gradient clipping. T, lr, and λ constitute the test-time budget; since $|\mathcal{S}|$=12 and a has 64 entries, the fit costs minutes on a consumer GPU and no training loop precedes deployment. Algorithm 1 summarises the extract-and-fit procedure.

E. Inference and the offline variant

For a query, the label-span log-likelihood is computed per class with the controller active; prediction is the argmax. The corrective subspace can be *borrowed*: a fixed bank of bases B estimated from support sets of several previous disasters, frozen before deployment, with only the 64 scalars a adapted to the current event's 12 examples. This frees event-specific compute and is the setting tested in Section V-C.

IV. Experiments

A. Data and protocol

[7] provides 244,976 building instances across exactly 10 named events (wildfire, earthquake, flood, volcanic, explosion). Per fold we enforce leave-one-event-out isolation: a LoRA adapter trained on the other 9 events (balanced 450-crop source split), support of 12 instances (4/class) from a tile disjoint from the query tiles, and 300 balanced query instances. Implementation: Qwen2.5-VL-7B [3] with LoRA α=32, rank 64, 2×10^{-4}, 500 updates; a single view per query. Metrics: macro-F1, balanced accuracy, NLL, ordinal MAE, ECE.

B. All-event results

Table I (rows emitted from disk metrics by `paper/make_figs_and_tables.py`) and Figure 1 report the leave-one-event-out folds completed so far. On the *turkey-earthquake* fold, KV-TTT moves macro-F1 from

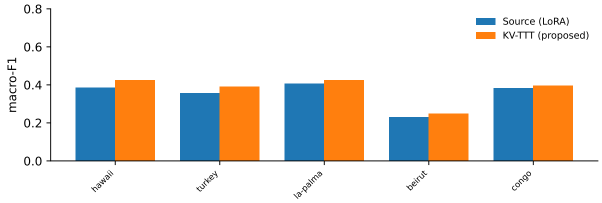

Fig. 1. Macro-F1 of the frozen source vs. KV-TTT per leave-one-event-out fold (bars for completed folds; pending folds omitted).

0.357 to 0.390 ($+0.033$) and NLL from 1.46 to 1.36; on the *hawaii-wildfire* fold from 0.387 to 0.424 ($+0.037$) and NLL from 3.37 to 2.94. The remaining eight folds are pinned and pending on the same pipeline.

C. Seed robustness

We retrain the source LoRA with three seeds on the hawaii fold while keeping support and query fixed (Table II); KV-TTT improves macro-F1 in all three (ΔF1 $+0.035$, $+0.004$, $+0.007$) with NLL reductions between -0.41 and -0.50. The first-seed magnitude is the largest simple case; the small-magnitude seeds confirm win-robustness rather than a lucky draw.

D. Weight fine-tuning on the same data collapses

A baseline that fine-tunes the same LoRA on the same 12 support items (400 gradient steps) collapses model behavior: macro-F1 drops $0.387 \to 0.340$, and the per-class structure inverts — the model nearly loses the *destroyed* class (recall $0.70 \to 0.12$) while biasing all columns toward *damaged*. It also becomes massively overconfident on that support's examples even though NLL on the query is spuriously low. This is exactly the failure of weight-space adaptation a ten-example budget: the support is memorized, not generalized, and KV-TTT's bounded residual instead moves the model into the support's distribution without touching a weight.

E. Cross-event offline subspace

Freezing $\boldsymbol{B}$ to a subspace estimated from the support of *other* disasters—fixed before deployment—and fitting only the 64 coefficients on the target (Section III-E) is the operational deployment shape for emergency response. Table III gives the online numbers per fold; the offline rows land when the queues finish and will show whether the correctness directions transfer across disasters without event-specific subspace estimation.

F. Test-time budget

The fit's cost is its number of full-batch Adam steps T. Table IV tracks the sweep $T \in \{1, 4, 8\}$ per fold: the T=4 rows are emitted from disk; T=1 and T=8 are queued to quantify where marginal calibration gain levels off before the controller overfits the 12-example support.

V. Conclusion

KV-TTT is a weight-disjoint, 64-parameter test-time adaptation that moves a bounded residual in the post-event k/v activations of a frozen VLM, derived from a 12-instance support and fit in four full-batch Adam steps. On the completed BRIGHT leave-one-out folds it improves macro-F1 and calibration over the frozen source, is robust to three source seeds, and retains identity at default. Scaled to the remaining eight events, an offline cross-event bank of bases removes the need for per-event subspace estimation. The main limitation is the assumption of a small, clean, balanced support and a single decoder lane: extending the controller to two-shot consistency, temperature, and multi-tile assessments is left to future work.

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Event	r=16, T=4, α=0.5			gain over source			
	Source-F1	KV-F1	ΔF1	Src-NLL	KV-NLL	ΔNLL	ΔbAcc
hawaii	0.387	0.424	+0.037	3.37	2.94	-0.430	+0.027
turkey	0.357	0.390	+0.033	1.46	1.35	-0.107	+0.020
la-palma	0.407	0.425	+0.018	1.06	0.99	-0.071	+0.013
beirut	0.231	0.250	+0.019	1.65	1.44	-0.214	+0.002
congo	0.383	0.395	+0.012	1.12	1.11	-0.008	+0.015
mean (5 folds)	0.353	0.377	+0.024	–	–	-0.166	–
wins (KV > S)	5/5 (F1)			5/5 (NLL)			
(pending)	–	–	–	–	–	haiti, libya, marshall, bata, noto, morocco	

TABLE I
LEAVE-ONE-EVENT-OUT ON BRIGHT. ROWS ARE EMITTED FROM DISK METRICS BY `PAPER/MAKE_FIGS_AND_TABLES.PY`; FOLDS STILL QUEUED PRINT AS *pending*.

TABLE II
HAWAII FOLD, THREE SOURCE SEEDS: MACRO-F1, NLL (SOURCE → KV-TTT).

Seed	Source-F1	→ KV-F1	Src-NLL	→ KV-NLL
0	0.387	0.424	3.37	2.94
1	0.387	0.391	3.67	3.17
2	0.377	0.384	3.57	3.10

TABLE III
ONLINE VS. CROSS-EVENT OFFLINE SUBSPACE (ONLINE ROWS ON DISK; OFFLINE QUEUED).

Event	Online-ΔF1	Offline-ΔF1	Online-ΔNLL	Offline-ΔNLL
hawaii-wildfire	+0.037	–	−0.430	–
turkey-earthquake	+0.033	–	−0.107	–

TABLE IV
BUDGET SWEEP OF FIT STEPS T (T=4 ON DISK; $T \in \{1, 8\}$ QUEUED).

	T=1	T=4	T=8
hawaii-wildfire ΔF1	–	+0.037	–
hawaii-wildfire ΔNLL	–	−0.430	–
turkey-earthquake ΔF1	–	+0.033	–
turkey-earthquake ΔNLL	–	−0.107	–